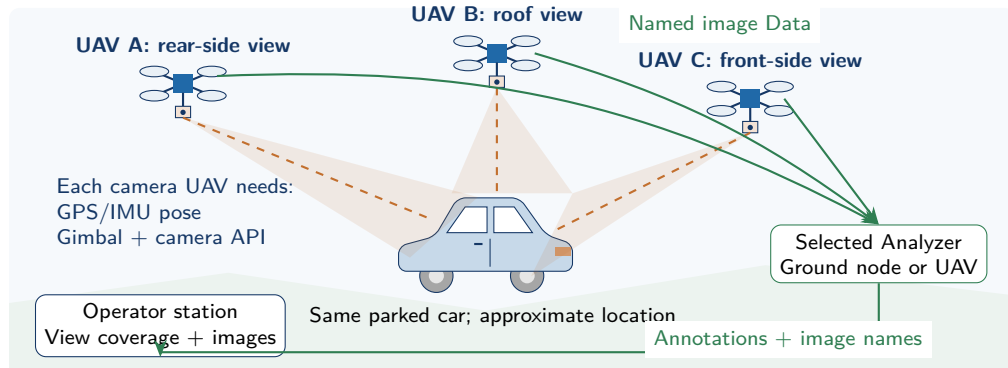


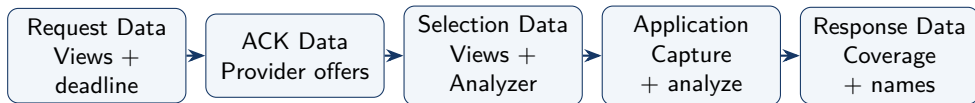
A Controlled Task: Collect Useful Views of One Car

Remote visual check: an operator requests rear, roof and front views of one parked car before a deadline.



Evidence: Proposed field test, not a completed flight experiment. Orange: views; green: named-data retrieval, not flight paths or unsolicited push.

Why This Is a Useful NDNF Test Case



- ▶ **Availability:** select camera and compute Providers at request time.
- ▶ **Dependencies:** retrieve selected views by name; report coverage and missing views.
- ▶ **Access rights:** authorize services and protect images across participating devices.

Key point

Compare against one UAV and ground compute; measure whether collaboration is worth its cost.

Evidence: Proposed mapping: NDNF handles invocation; the application validates capture and same-car association.

Three Gaps Between Simulation and the Field

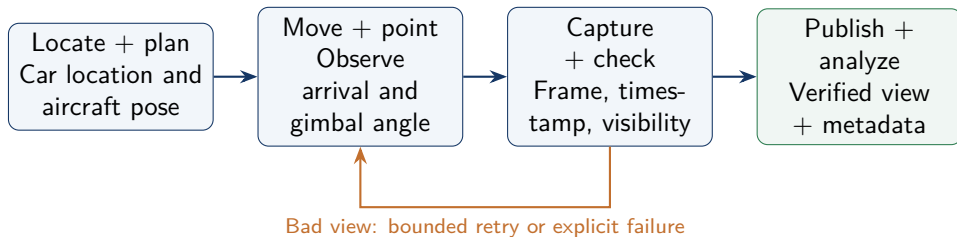
Gap	Simplified example	Real deployment needs
Position / pointing	Known coordinates; assigned views	GPS + IMU, camera/gimbal calibration, coordinate frames, synchronized pose and exposure
Useful images	Car assumed visible	Real control/capture APIs, feedback, blur/occlusion checks and bounded retries
Multi-view result	Images supplied to a detector	Same-car association, requested-view coverage and independent image labels

Key point

Start with verified, complementary images. Damage diagnosis and 3-D reconstruction require separate algorithms and validation.

Evidence: Candidate tools: MAVSDK control/telemetry; OpenCV calibration/geometry. Hardware support and integration remain to be verified.

Close One Physical Loop Before Adding More UAVs



- ▶ A camera command acknowledgement is not capture success: check the image and measured pose.
- ▶ Match each image to the same car and requested view; reject stale or unusable frames.
- ▶ Keep stabilization and safety rejection onboard, independent of network delays.

Evidence: Proposed feedback loop: reject stale pose, missing frames and exhausted deadlines.

Test Camera Benefit and Framework Benefit Separately

Question	Comparison	Measure
Are extra UAVs useful?	One UAV visits all views vs. multiple UAVs	Useful-view coverage, completion time, total energy
Does selection help?	Fixed assignments vs. runtime selection; same fleet	Completion under Provider loss, deadline misses, network bytes

Route: camera bench → flight simulation → one real UAV → multiple UAVs.

Key point

Fix the scene, views and deadline. Label visibility independently; include failed runs.

Evidence: Planned ablation, no measured advantage. Match cameras, models and workload; include ground compute. This is not a comparison with other frameworks.

Candidate Hardware for the First Camera Bench Test

SIYI A8 mini

SIYI



- ▶ Camera + three-axis gimbal in one unit.
- ▶ Ethernet UDP / UART control; Ethernet video.

Gremsy Pixy U



- ▶ Gimbal only; add a compatible camera.
- ▶ SDK / MAVLink control; verify camera support.

First bench test: command an angle → read actual attitude → capture the parked car.

Evidence: Product images: SIYI official store and Gremsy official site. Images not to scale. Candidate hardware; verify power, payload and mounting before selection.